

COM S 472 – Lecture 1

Introduction to Artificial Intelligence

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Lecture: MWF, 3:20 PM – 4:10 PM, 1104 Gilman

Recitation: W 8.50 AM - 9.40 AM, 1104 Gilman

Course syllabus can be found on canvas

Prereq: COMS 3110; STAT 3300 or STAT 3050 or STAT 3410 or STAT 3470; MATH 1650; ENGL 2500

Description:

- Basic principles of Artificial Intelligence, including problem solving, knowledge representation, reasoning, decision making, planning, and learning
- Applications of AI, e.g., game playing, text classification, medical diagnosis, data mining, information retrievalDesign and implementation of intelligent agents

Programming Language: java

Resources

Canvas: <https://canvas.iastate.edu/courses/126069>

Piazza: <https://piazza.com/iastate/spring2026/s2026coms472001>

Textbook: S. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach* (4th edition). Pearson Education, Inc., 2020.

Exercises: <http://aima.cs.berkeley.edu/>

Assessments (COMS 4720, COMS 5720)

- Homework assignments (20%, 20%)
 - Include theoretical questions and implementation tasks
 - Some assignments include additional work for COM S 5720
- Midterm 1 (22%, 22%)
- Midterm 2 (25%, 25%)
- Final exam (33%, 25%)
- Technical survey (0%, 8%)



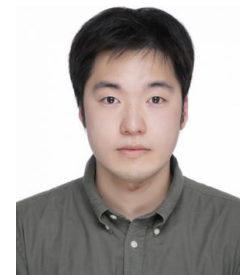
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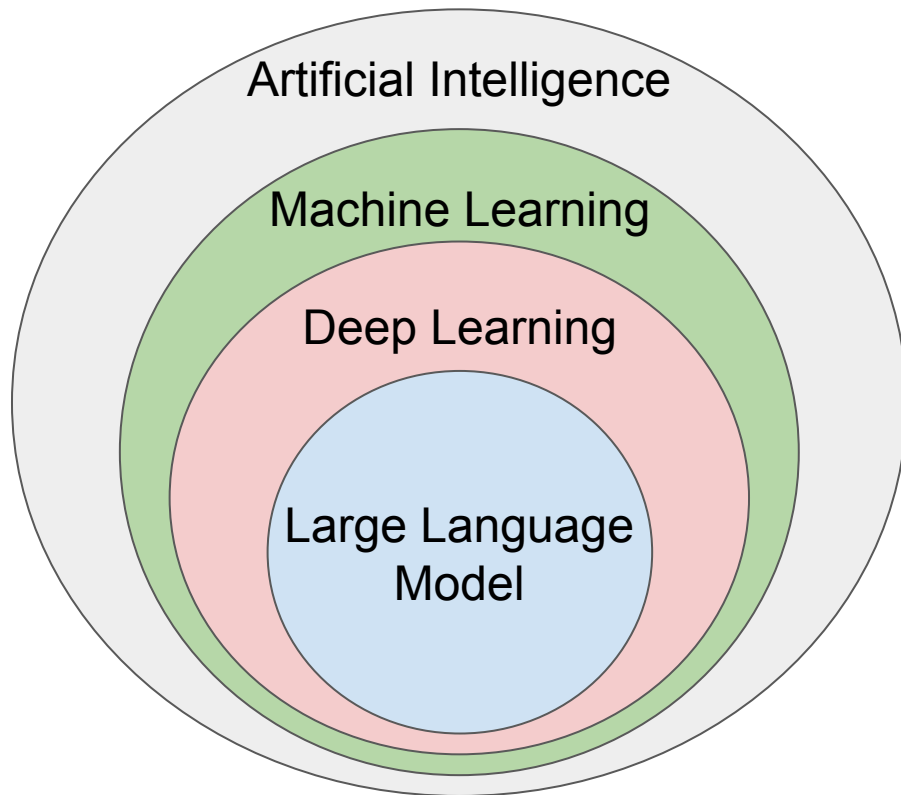


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What is AI?



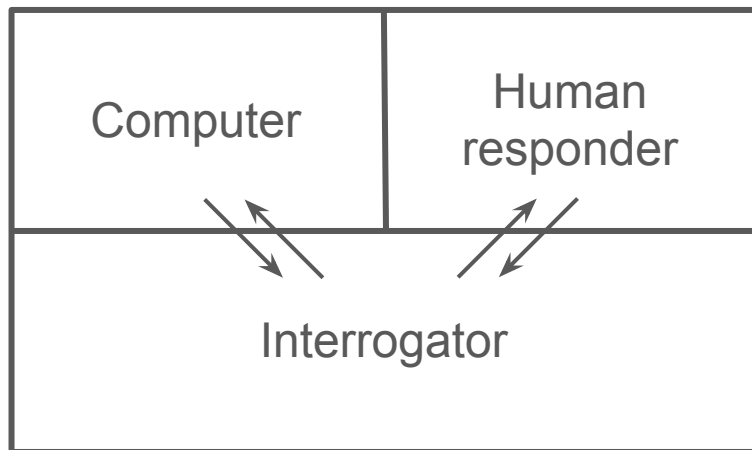
- AI is a broad and evolving field
- “Intelligence” is hard to define precisely
- There is no single agreed-upon definition of AI
- AI broadly refers to systems designed to perform tasks that normally require human intelligence

AI definitions: Thinking vs. Acting and Humanly vs. Rationally

	Human	Rational
Internal thought process	Thinking humanly	Thinking rationally
External behavior	Acting humanly	Acting rationally

Acting Humanly: Turing Test

Goal: Make machines behave like humans



Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, 59, 433–460

- 3 players communicate via text-only interfaces
 - a computer
 - a human respondent
 - a human interrogator
- A human interrogator poses some written questions to another human and a computer
- The computer passes the test if the interrogator cannot tell whether the written responses come from the human responder or the computer.
- Focus on imitation and ignore other aspects of intelligence

Capabilities required to pass the Turing test

- **Natural language processing (NLP)** to communicate in a human language
- **Knowledge representation** to store what it knows
- **Automated reasoning** to answer questions and to draw new conclusions
- **Machine learning (ML)** to adapt to new circumstances and to detect patterns

Total Turing test requires interaction with the real world

- **Computer vision (CV) and speech recognition** to perceive the world;
- **Robotics** to manipulate objects and move around in the world

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Thinking Humanly: Cognitive Modeling

Goal: Model human thought processes

- Introspection: catching our own thoughts as they go by;
- Psychological experiments: observing a person in action;
- Brain imaging: observing the brain in action

Cognitive science constructs theories of the human mind by combining

- Computer models from AI
- Experimental techniques from psychology to mind.

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Thinking Rationally: “Laws of Thought”

Goal: Use logic and reasoning to make decisions

- Aristotle’s “laws of thought”
- Focus on correct inferences

Limitations

- Requires knowledge of the world that is certain
 - Use theory of probability to allow rigorous reasoning with uncertain information
- Does not generate intelligent *behavior*

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The focus of this course

Acting Rationally: Rational Agent

Goal: Build agents that do “the right thing”

- Most practical & widely used definition today
 - More general than the “laws of thought” approach
 - Mathematically well defined and completely general
- An agent is an entity that perceives and acts
 - operate autonomously
 - perceive their environment
 - persist over a prolonged time period
 - adapt to change
 - create and pursue goals
- A rational agent is one that acts to achieve the best (expected) outcome

Rational Agent

- Built on logical foundations
 - formed definite plans to achieve specific goals.
- Incorporate probability theory and machine learning to enable decision-making under uncertainty to attain the best expected outcome.
- “The right thing” is defined by the objective that we provide to the agent
 - “Standard model”
 - Also used in control theory, operations research, statistics, and economics
- Challenges
 - Value alignment problem: specifying the objective completely and correctly (i.e., aligned with those of human)

Examples

